

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 03



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Recap of Previous Lecture



Topic

Venn diagram

Topic

Set operations and properties of set operations

Topic

Multi-set

Topics to be Covered



✓
Topic

Cartesian product

✓
Topic

Relation

✓
Topic

Diagonal relation, Reflexive and irreflexive relation

✓
Topic

Symmetric, anti-symmetric and asymmetric relation

✓
Topic

Transitive relation



Topic : Multi-set

- An unordered collection of objects/elements in which an element may appear more than once is called a multiset.

eg: Multiset $A = \{a, b, b, a, a, c, d, c\}$

Size of multiset $A = 8$

$= \{3.a, 2.b, 2.c, 1.d\}$

These numbers will represent the multiplicity of the corresponding elements

$= 3 + 2 + 2 + 1 = 8$

Let multiset $P = \{m_1 \cdot a_1, m_2 \cdot a_2, m_3 \cdot a_3, \dots, m_k \cdot a_k\}$
where ' m_i ' is the multiplicity of element a_i .

And
let multiset $Q = \{n_1 \cdot a_1, n_2 \cdot a_2, n_3 \cdot a_3, \dots, n_k \cdot a_k\}$
where ' n_i ' is the multiplicity of element a_i .

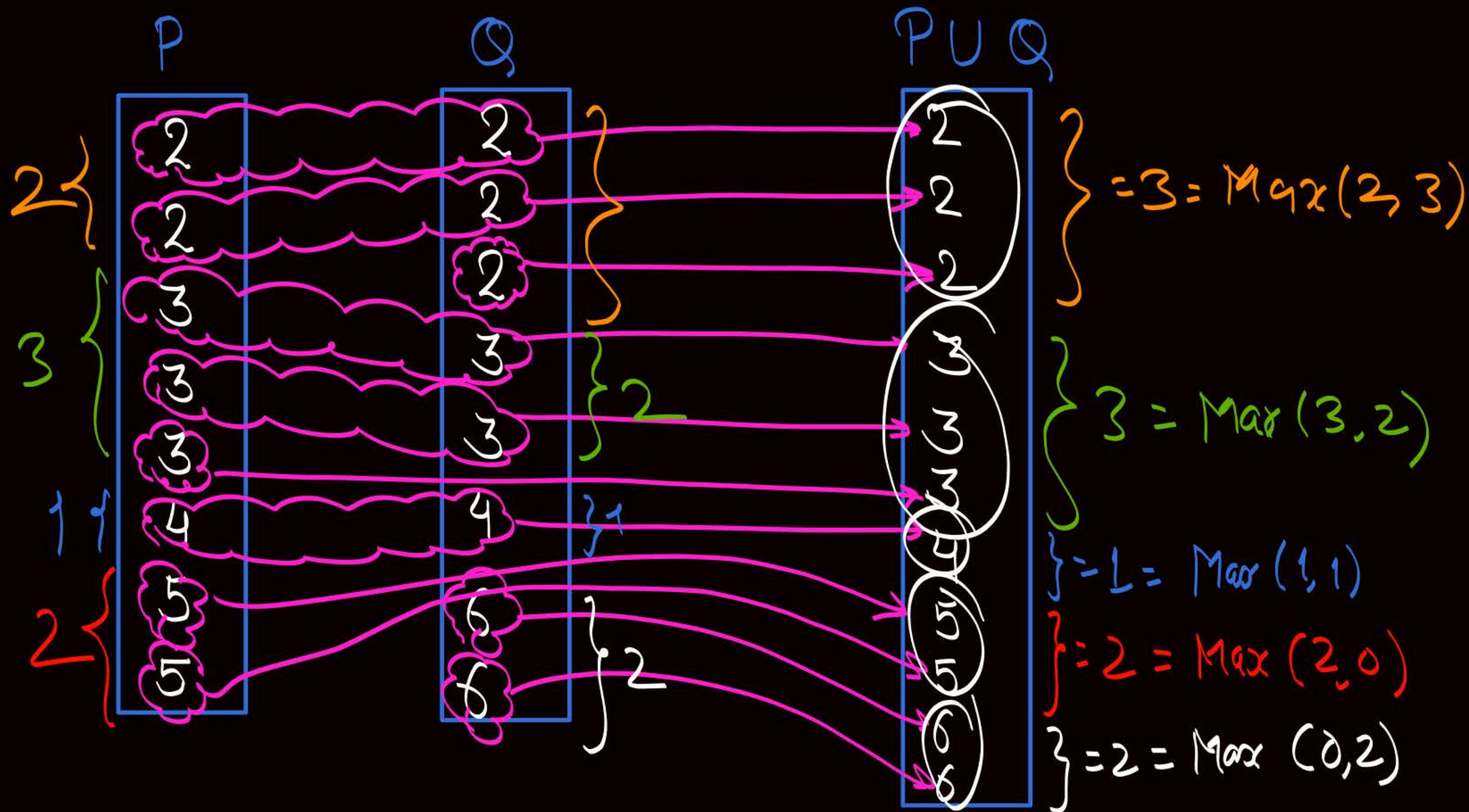
Find multiplicity of a_i in

(i) $P \cup Q$

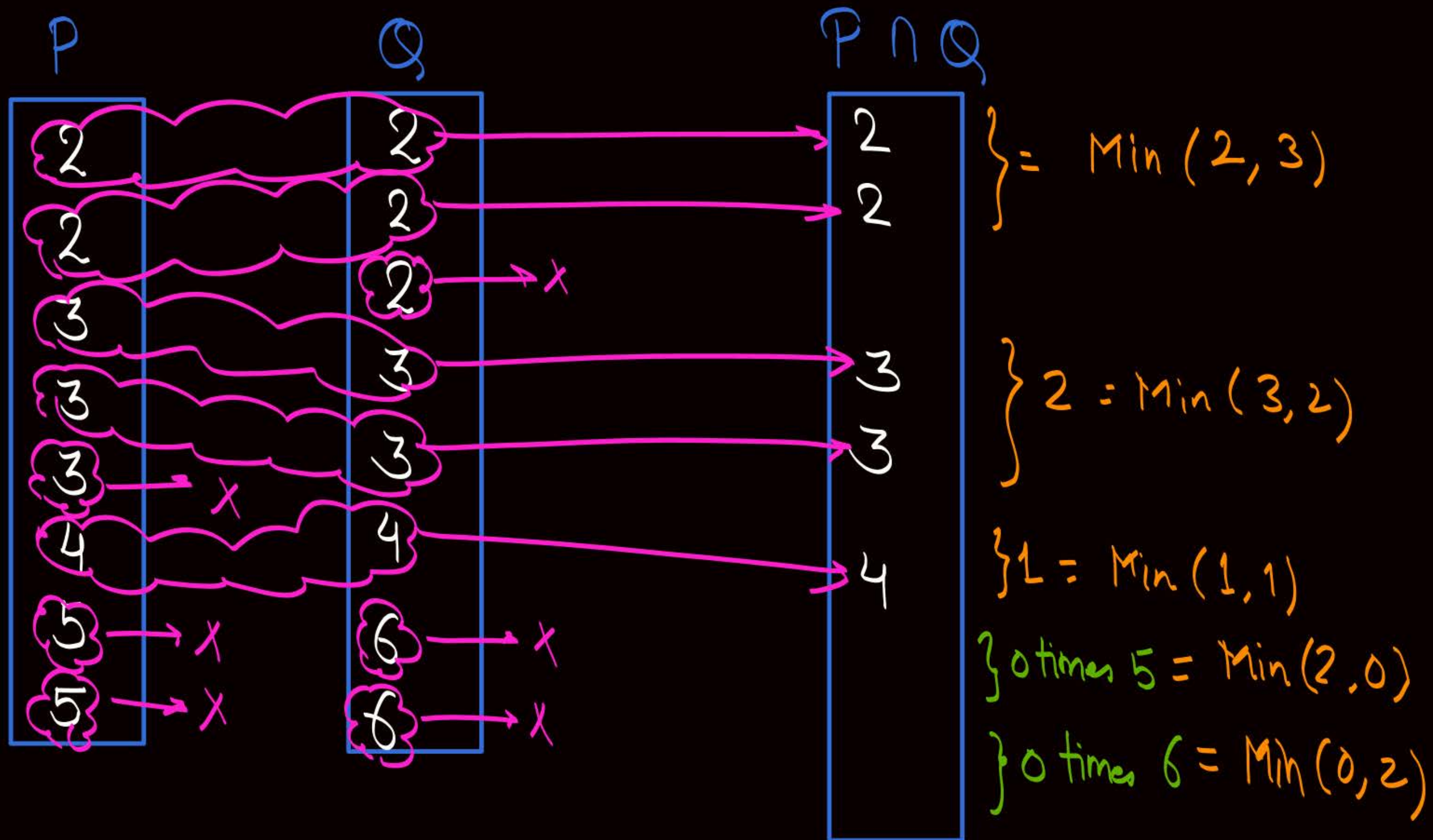
(ii) $P \cap Q$

(iii) $P - Q$

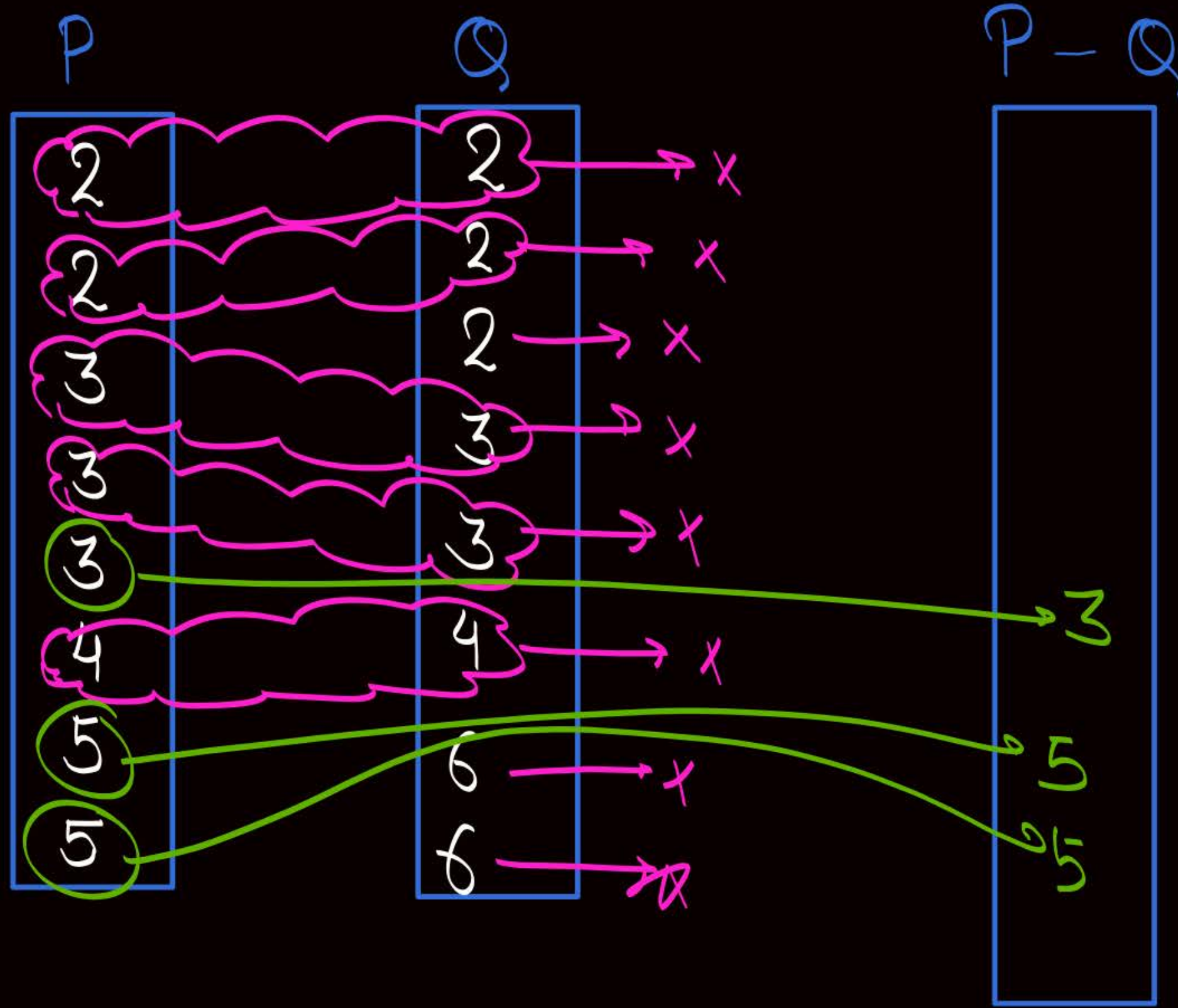
Multiset $P \cup$ Multiset Q



Multiset $P \cap$ Multiset Q



Multiset P — Multiset Q



Let multiset $P = \{m_1.a_1, m_2.a_2, m_3.a_3, \dots, m_k.a_k\}$
Where ' m_i ' is the multiplicity of element a_i .

And
let multiset $Q = \{n_1.a_1, n_2.a_2, n_3.a_3, \dots, n_k.a_k\}$
Where ' n_i ' is the multiplicity of element a_i .

Find multiplicity of a_i in

(i) $P \cup Q = \text{multiplicity of } a_i \text{ in } P \cup Q = \max(m_i, n_i)$

(ii) $P \cap Q = \text{multiplicity of } a_i \text{ in } P \cap Q = \min(m_i, n_i)$

(iii) $P - Q = \text{multiplicity of } a_i \text{ in } P - Q = \begin{cases} (m_i - n_i) & \text{if } m_i > n_i \\ 0 & \text{otherwise} \end{cases}$

Q: Let $A = \{1, 2, 3, 4, \dots, n\}$

How many different multisets can be created using the element of set A. { Any element of set A
Can be taken any no. of times }

$$M_1 = \{1\}$$

$$M_2 = \{1, 1\}$$

$$M_3 = \{1, 1, 1\}$$

$\vdots$

$$M_{100} = \{1, 1, 1, \dots, 1\}$$

100 times '1'

$\vdots$

$$M_\infty = \{1, 1, \dots\}$$

infinite times 1

If we do not restrict the size of multisets then we can create infinite multisets using a single element

$\therefore$ Ans = infinite

Q: Let $A = \{1, 2, 3, 4, \dots, n\}$
How many different multisets of size = 4 can be
formulated using the elements of set A, such that
at least one element of set A appears exactly
twice in each such multiset.

Based on the given statement only two types of multisets
 are possible.

$A = \{1, 2, 3, \dots, n\}$

Type-1 $\{a, a, b, c\}$

$\{a, b, a, c\}$

$${}^nC_1 \times \frac{{}^{(n-1)}C_1 \times {}^{(n-2)}C_1}{2!} = \frac{n(n-1)(n-2)}{2}$$

$$({}^nC_1) \times ({}^{(n-1)}C_2) = \frac{n(n-1)(n-2)}{2}$$

$\{a, a, b, b\}$ Type-2

$\{a, b, b, a\}$

$$\frac{{}^nC_1 \times {}^{(n-1)}C_1}{2} = \frac{n(n-1)}{2}$$

$$({}^nC_2 \times 1) = \frac{n(n-1)}{2}$$

a, b, c

a, a, b, c

b, b, a, c

c, c, a, b

$${}^nC_3 \times 3 \times 1 \times 1 = \frac{n(n-1)(n-2)}{3 \cdot 2 \cdot 1}$$

$$= \frac{n(n-1)(n-2)}{2}$$

Q: Let $A = \{1, 2, 3, 4, \dots, n\}$
How many different multisets of size = 4 can be formulated using the elements of set A, such that at least one element of set A appears exactly twice in each such multiset.

* Required multiset may be of Type-1 or Type-2

∴ Total no. of Multisets of required type =

No. of multisets using Type-1 + No. of multisets using Type-2

$$= \frac{n(n-1)(n-2)}{2} + \frac{n(n-1)}{2} = \boxed{\frac{n(n-1)^2}{2}}$$

Today's Topic

H.W.

Q:

Let $A = \{1, 2, 3, 4, \dots, n\}$

How many different multisets of 'size = 4' can be formulated using the elements of set A.

Possible types of multisets are

$$\textcircled{1} \{a, b, c, d\} \Rightarrow {}^nC_4 = \frac{n(n-1)(n-2)(n-3)}{4!}$$

$$\textcircled{2} \{a, a, b, c\} \Rightarrow {}^nC_3 * {}^3C_1 * 1 = \frac{n(n-1)(n-2)}{2}$$

$$\textcircled{3} \{a, a, b, b\} \Rightarrow {}^nC_2 = \frac{n(n-1)}{2}$$

$$\textcircled{4} \{a, a, a, b\} = {}^nC_2 * {}^2C_1 * 1 = n(n-1)$$

$$\textcircled{5} \{a, a, a, a\} = {}^nC_1 * 1 = n$$

Total no. of multisets of size '4' = $\left(\frac{n(n-1)(n-2)(n-3)}{24} \right)$

$$+ \left(\frac{n(n-1)(n-2)}{2} \right) + \left(\frac{n(n-1)}{2} \right)$$

$$+ n(n-1) + n$$

Relation

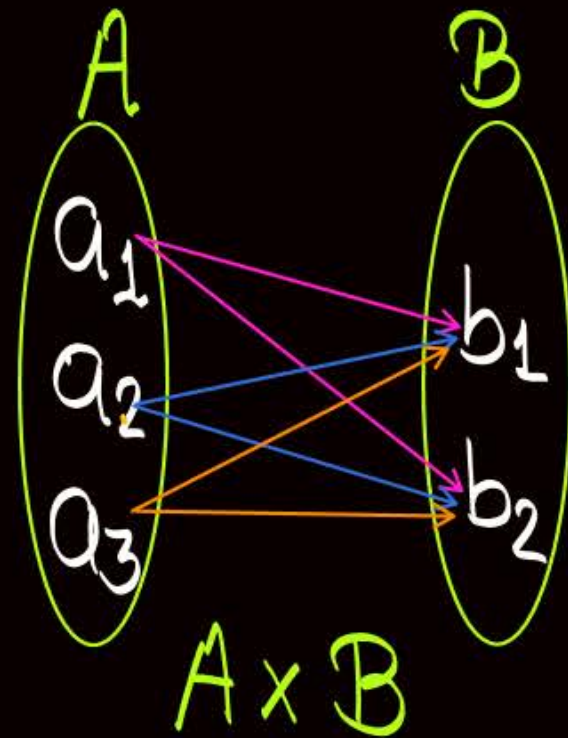
★ Cartesian Product / Cross Product

Let A and B are two sets,
Cartesian product of A and B is denoted by ' $A \times B$ '
and it is defined as

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Order pairs
ie, Order matter
 $(a, b) \neq (b, a)$

eg. Let $A = \{a_1, a_2, a_3\}$
 $B = \{b_1, b_2\}$



$$\underline{A \times B} = \{ (a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2) \}$$

$$B \times A = \{ (b_1, a_1), (b_1, a_2), (b_1, a_3), (b_2, a_1), (b_2, a_2), (b_2, a_3) \}$$

We know, $(a_1, b_1) \neq (b_1, a_1)$ & in general $A \times B \neq B \times A$

In general,

$$A \times B \neq B \times A$$

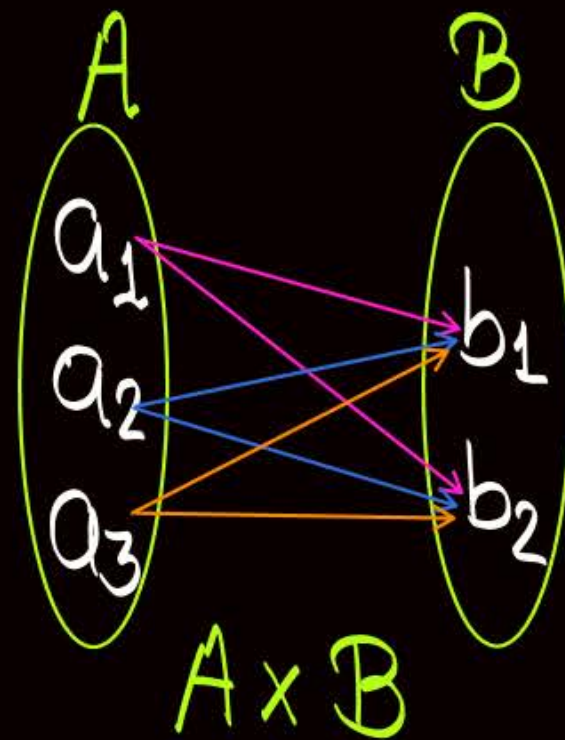
- " $A \times B = B \times A$ " if and only if
either $A = B$
or at least one of set A or B is an Empty Set.

If at least one of A or B is an
Empty set, then

$$A \times B = B \times A = \{ \}$$

Result will be an Empty Set.

eg. Let $A = \{a_1, a_2, a_3\}$
 $B = \{b_1, b_2\}$



$$A \times B = \{ (a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2) \}$$

Note: We can observe that in $A \times B$ Every element of set A relates with every element of set B.

$\therefore$ If $|A| = m$ & $|B| = n$
then $|A \times B| = m \cdot n = |A| \cdot |B|$



Topic : Relation

A relation from set A to set B defines that how exactly the elements of set A relates with the elements of set B .

- In Cartesian product ' $A \times B$ ' Every element of set A relates with every element of set B .
i.e. ' $A \times B$ ' is the universal set w.r.t. relations from set A to set B .

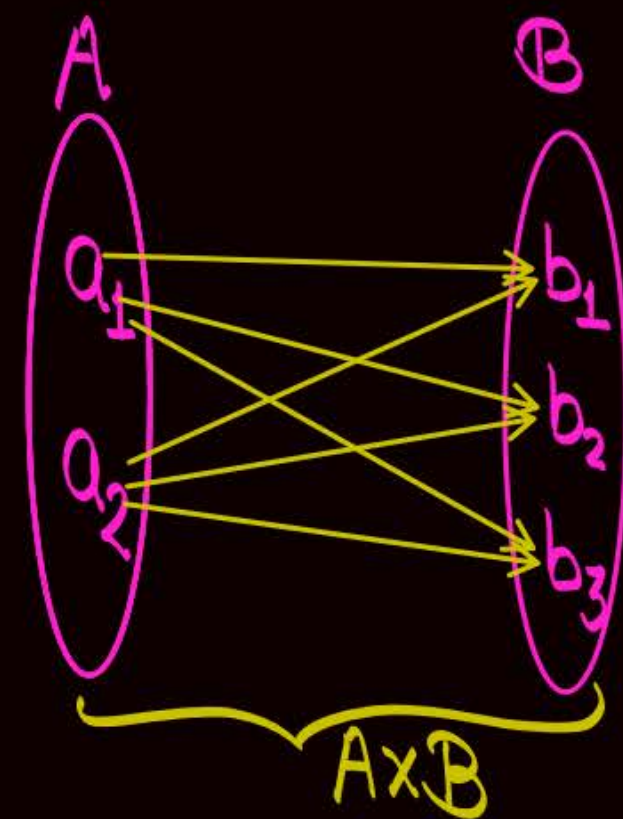
Any relation from set A to set B is a subset of $A \times B$

eg:-

$$A = \{a_1, a_2\}$$

$$B = \{b_1, b_2, b_3\}$$

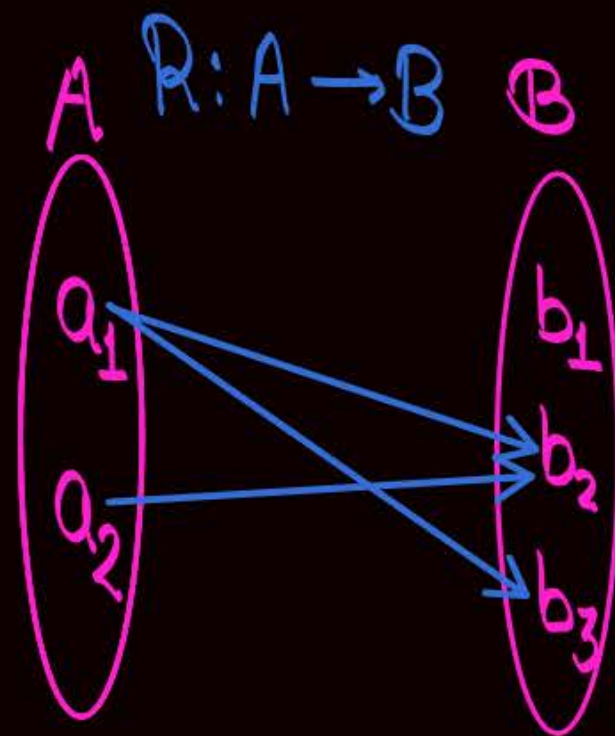
$$A \times B = \{(a_1, b_1), (a_1, b_2), (a_1, b_3), (a_2, b_1), (a_2, b_2), (a_2, b_3)\}$$



Let relation R denotes a relation from set A to set B defined by.

$$R = \{(a_1, b_2), (a_1, b_3), (a_2, b_2)\}$$

We can observe that all order pairs of relation R are member of $A \times B$ $\therefore R$ is a subset of $A \times B$



* Number of different relations possible from set A to set B

Let A and B are two sets

such that $|A|=m$ and $|B|=n$, then

* How many different relations are possible from set A to set B?

Every subset of $A \times B$ will be a relation from A to B

$$\begin{aligned} \therefore \text{Total no. of relations from A to B} &= \text{No. of subsets of } A \times B \\ &= 2^{|A \times B|} = 2^{|A| \cdot |B|} \\ &= 2^{m \cdot n} \end{aligned}$$

✓ One subset of ' $A \times B$ ' is
Empty set.

This Empty subset of $A \times B$
is also a relation from set A
to set B, and it is called
"Empty relation" from set A to set B.



Topic : Relation

A relation from set A to set A itself is called a relation on set A

⊛ Every relation on set A is a subset of " $A \times A$ "

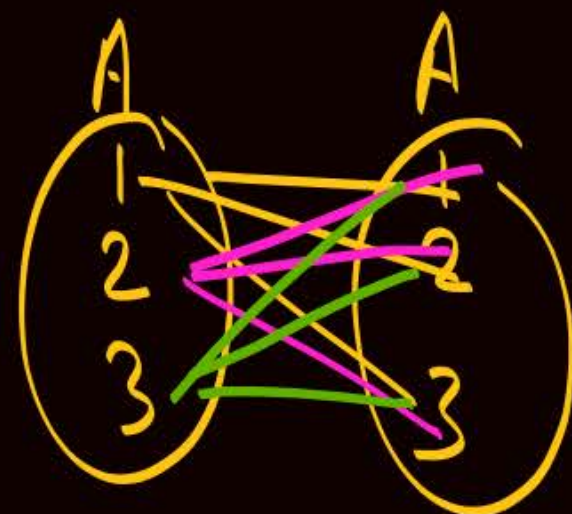
* If $|A| = n$, then

$$\begin{aligned} \text{Number of relations possible on set } A &= \text{No. of subsets of } A \times A \\ &= 2^{|A \times A|} = 2^{|A| \cdot |A|} \\ &= 2^{n \cdot n} = 2^{(n^2)} \end{aligned}$$

$$A = \{1, 2, 3\}$$

$$A \times A = \left\{ \begin{array}{l} (1,1), (1,2), (1,3) \\ (2,1), (2,2), (2,3) \\ (3,1), (3,2), (3,3) \end{array} \right.$$

$$(3,1), (3,2), (3,3)$$



$$\Rightarrow \text{No. of subsets} = 2^9 = 2^{(3^2)}$$



Topic : Types of Relations



All these relations are defined on same set, i.e., from a set A to set A itself

- (1) Diagonal Relation (Identity Relation)
- (2) Reflexive Relation
- (3) Irreflexive Relation
- (4) Symmetric Relation
- (5) Anti-symmetric Relation
- (6) Asymmetric Relation
- (7) Transitive Relation

- (8) Complement of a relation
- (9) Inverse of a relation
- (10) Composite of two relations



Topic : Diagonal Relation

(Identity Relation)



A diagonal relation on set A (i.e. identity relation on set A)
Can be denoted by Δ_A (or I_A) and it is
defined as,

"It is the definition of set"

$$\Delta_A = \{ (a, a) \mid a \in A \}$$

eg Let $A = \{1, 2, 3\}$

$R_1 = \{ (1, 1), (2, 2) \}$ $\Rightarrow 3 \in A$, but $(3, 3) \notin R_1$ $\therefore R_1$ is not a diagonal relⁿ on set A

$R_2 = \{ (1, 1), (2, 2), (3, 3), (2, 3) \}$ $\Rightarrow 2 \neq 3$, but $(2, 3) \in R_2$ $\therefore R_2$ is not a diagonal relation

$R_3 = \{ (1, 1), (2, 2), (3, 3) \}$ is a diagonal relⁿ on set A

Note:-

on any given set A , Exactly "one"
diagonal relation is possible



Note:-

Cardinality of diagonal relation on set $A = |A|$

$$\text{Let } A = \{1, 2, 3, 4, \dots, n\}$$

$$A \times A = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), \dots, (1,n) \\ (2,1), (2,2), (2,3), \dots, (2,n) \\ (3,1), (3,2), (3,3), \dots, (3,n) \\ \vdots \\ (n,1), (n,2), (n,3), \dots, (n,n) \end{array} \right\}$$

Let $A = \{1, 2, 3, 4, \dots, n\}$

These order
Pairs are
called
diagonal order
pairs

$A \times A =$

$(1,1), (1,2), (1,3), \dots, (1,n)$
 $(2,1), (2,2), (2,3), \dots, (2,n)$
 $(3,1), (3,2), (3,3), \dots, (3,n)$
 $\vdots$
 $(n,1), (n,2), (n,3), \dots, (n,n)$

All other order
Pairs are called
non-diagonal
order Pairs.

• If $|A| = n$, then

(i) Total no. of order
Pairs in $A \times A = n^2$

(ii) No. of diagonal
order Pairs in $A \times A = n$

(iii) No. of non-diagonal
order Pairs in $A \times A = (n^2 - n)$

Note:- Let 'R' is any relation on set A,
and $a, b \in A$,

then $(a, b) \in R$ if and only if (a relates with b
w.r.t relation R)

↓
i.e. $a R b$

$a \not R b$: "a does not relate with b"
w.r.t. relation R, then $(a, b) \notin R$



Topic : Reflexive Relation

A relation R on set ' A ' is called a reflexive relation if and only if

It is a constraint for relation R to be called a reflexive relation

$$\left\{ \begin{array}{l} a^R a, \forall a \in A \\ \text{ie. } (a, a) \in R, \forall a \in A \end{array} \right.$$

" $a^R b$ " is read as
' a relates with b ' w.r.t.
relation R .
ie. $(a, b) \in R$

" $a \cancel{R} b$ " is read as
 a does not relate with b
w.r.t. relation R .
ie. $(a, b) \notin R$

eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{(1,1), (2,2), (3,3)\}$: R_1 is a diagonal relation on set A as well as a reflexive relⁿ on set A

② $R_2 = \{(1,1), (2,2), (3,1), (3,3)\}$ } R_2 is not a diagonal relation but $\forall a \in A, (a,a) \in R_2 \therefore R_2$ is reflexive

③ $R_3 = \{(1,1), (2,2)\}$ $3 \in A$, but $(3,3) \notin R_3 \therefore R_3$ is not reflexive

④ $R_4 = \{(1,2), (2,3), (3,1)\}$ R_4 is not reflexive

Note:- ① Every diagonal relation on set A is also a reflexive relation on set A , but every reflexive relation on set A need not be a diagonal relation on set A .

② Diagonal relation on set A is the smallest reflexive relation on set A .

③ Largest reflexive relation on set A is " $A \times A$ "

Q: Let $A = \{1, 2, 3, \dots, n\}$

How many reflexive relation are possible on set A
Where $|A| = n$.

Reflexive Relation = (All 'n' out of 'n' diagonal order pairs must be present)

And

(Along with all diagonal order pairs, any number of non-diagonal order pairs may be present)

$$= nC_n * \left\{ \binom{n^2-n}{0} + \binom{n^2-n}{1} + \binom{n^2-n}{2} + \dots + \binom{n^2-n}{n^2-n} \right\}$$
$$= 1 * \{ 2^{n^2-n} \}$$

$$= \boxed{2^{n^2-n}}$$

Let $A = \{1, 2, 3, 4, \dots, n\}$

$$A \times A = \left\{ \begin{array}{l} (1, 1), (1, 2), (1, 3), \dots, (1, n) \\ (2, 1), (2, 2), (2, 3), \dots, (2, n) \\ (3, 1), (3, 2), (3, 3), \dots, (3, n) \\ \vdots \\ (n, 1), (n, 2), (n, 3), \dots, (n, n) \end{array} \right\}$$



Topic : Irreflexive Relation

A relation R on set A is called irreflexive relation if and only if

$$a \not R a, \forall a \in A$$

ie $(a, a) \notin R, \forall a \in A$

} i.e. no diagonal order pair should be present in the relation

eg: Let, $A = \{1, 2, 3\}$

① $R_1 = \{ \}$ Empty relation is the smallest irreflexive relation on any set A
Empty relⁿ

② $R_2 = \{(1,2), (2,3), (3,1)\}$ R_2 is an irreflexive relⁿ

③ $R_3 = \{(1,2), (2,1), (2,2), (3,2)\}$
 $(2,2) \in R_3$
 $\therefore R_3$ is not an irreflexive relation.
 $(1,1) \notin R_3$ & $(3,3) \notin R_3$
 $\therefore R_3$ is not reflexive as well.

Q: Let $A = \{1, 2, 3, \dots, n\}$

How many "irreflexive relation" are possible on set A
where $|A| = n$.

Irreflexive relation = ('0' out of 'n' diagonal order Pair should be present) & (any no. of non-diagonal order pairs out of $(n^2 - n)$ may be present)

$$= nC_0 * (n^2 - nC_0 + n^2 - nC_1 + \dots + n^2 - nC_{n^2 - n})$$
$$= 1 * 2^{n^2 - n}$$

$$= \boxed{2^{n^2 - n}}$$



Topic : Symmetric Relation

A relation R on set A is called a symmetric relation if and only if

$$\text{if } aRb \text{ then } bRa \quad \forall a, b \in A$$

i.e, if $(a, b) \in R$ then $(b, a) \in R \quad \forall a, b \in A$

eg: Let, $A = \{1, 2, 3\}$

* ① $R_1 = \{ \}$ Empty relation is the smallest symmetric relation

② $R_2 = \{(1, 1)\}$ it is symmetric relation { Presence or Absence of diagonal order pair will never create any problem w.r.t. Symmetry

③ $R_3 = \{(1, 2), (2, 1), (2, 2)\}$ it is symmetric relation

④ $R_4 = \{(1, 1), (2, 3), (3, 1), (3, 2)\}$ but $(1, 3) \notin R_4$ so Not R_4 is not a symmetric Relⁿ

→ Empty relation on any set A is the smallest symmetric relation

→ ' $A \times A$ ' is the largest symmetric relation on any set A .

Q: Let $A = \{1, 2, 3, \dots, n\}$

How many "symmetric relation" are possible on set A
Where $|A| = n$.

Number of symmetric relation = $\left(\begin{array}{l} \text{Any no. of} \\ \text{diagonal order} \\ \text{Pairs may be} \\ \text{Present} \end{array} \right)$ and $\left(\begin{array}{l} \text{Possible combinations} \\ \text{w.r.t. non-diagonal} \\ \text{order pairs} \end{array} \right)$

$\downarrow$

$\left(2^n \right) * \left(? \right)$

→ in non-diagonal order pair (a, b) , $a \neq b$

• i.e. for non-diagonal order pair we need two distinct elements $a \neq b$

• Want two distinct elements $a \neq b$
What are the different combinations of order pairs that may be present in a relation

(i)

$(a, b), (b, a)$

$aRb \neq bRa$

(ii)

$(a, b) \cancel{(b, a)}$

$aRb \neq bRa$

(iii)

$\cancel{(a, b)} (b, a)$

$aRb \neq bRa$

(iv)

$\cancel{(a, b)} \cancel{(b, a)}$

$aRb \neq bRa$

Approach w.r.t. non-diagonal order pairs

→ In a non-diagonal order pair (a, b) , we know that $a \neq b$

if $a = b$, then (a, b) is a diagonal order pair.

- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

→ W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation.

Case (i)	Case (ii)	Case (iii)	Case (iv)
both $(a, b), (b, a)$ are present	(a, b) is present but (b, a) is not present	(a, b) is not present but (b, a) is present	Both (a, b) and (b, a) are not present

There are the only possible combinations of order pairs in a relation w.r.t. two distinct elements a & b

Approach w.r.t. non-diagonal order pairs

→ In a non-diagonal order pair (a, b) , we know that $a \neq b$

if $a = b$, then (a, b) is a diagonal order pair.

• For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

→ W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation.

Case (i)

both $(a, b), (b, a)$
are present

Allowed in
Symmetric relⁿ

Case (ii)

(a, b) is present
but (b, a) is not present

Not Allowed in
Symmetric relⁿ

Case (iii)

(a, b) is not present
but (b, a) is present

Not Allowed in
Symmetric relⁿ

Case (iv)

Both (a, b) and (b, a)
are not present

Allowed in
Symmetric relⁿ

Let $A = \{a, b, c, d\}$

Out of '4' we can choose two distinct elements in ${}^4C_2 = 6$ ways

Total possible combinations w.r.t. non-diagonal order pairs in a symmetric Relⁿ

$a \& b$

↓
'2'

Choices w.r.t.
 $a \& b$

And
✗

$a \& c$

↓
2

Order
✗

$a \& d$

↓
2

Order
✗

$b \& c$

↓
2

Order
✓

$b \& d$

↓
2

Order
✗

$c \& d$

↓
2

$$= 2^6 = 2^{({}^4C_2)}$$

ie. $(a, b), (b, a)$ both present
or
 $(a, b), (b, a)$ both absent

Q: Let $A = \{1, 2, 3, \dots, n\}$
 How many "symmetric relation" are possible on set A
 where $|A| = n$.

Number of symmetric relation = (Any no. of diagonal order Pairs may be present) and (Possible combinations w.r.t. non-diagonal order pairs)

$$= (2^n) * \frac{n(n-1)}{2}$$

$$= 2^n \cdot 2^{\frac{n(n-1)}{2}}$$

$$= 2^n \cdot 2^{\frac{n^2-n}{2}} = 2^{\frac{n^2+n}{2}}$$

$$= 2^{\frac{n(n+1)}{2}}$$

We can choose two distinct elements out of 'n' in nC_2 ways
 for each pair of distinct elements we have '2' choices of order pairs that are allowed in a symmetric relⁿ



Topic : Anti-Symmetric Relation

A relation R on set A is called an anti-symmetric relation if and only if
if $(a^R b \text{ and } b^R a)$ then $(a=b)$, $\forall a, b \in A$

i.e. if $((a,b) \in R \text{ and } (b,a) \in R)$ then $(a=b)$, $\forall a, b \in A$

∴ diagonal order pairs
will never create
any problem w.r.t.
Anti-symmetric
Property

' $a=b$ ' means
diagonal order pairs
↓
i.e. w.r.t. diagonal order
pairs "if-then" statement
will always be true

If $a=b$, then "then" condⁿ itself
of "if-then" statement is true.
And hence when " $a=b$ "
"if-then" statement will always be true

Note :-

In an antisymmetric relation there will be no problem because of presence or absence of any diagonal order pair.

But if $a \neq b$, i.e. when "then" statement is false
then for "if-then" statement to be true
i.e. In an anti-symmetric relation ^{our} "if" statement itself must be false

if $a \neq b$, then

either (i) $(a, b) \notin R$

or (ii) $(b, a) \notin R$

or (iii) $(a, b) \notin R$ and $(b, a) \notin R$

eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{ \}$ Empty relation is the smallest anti-symmetric relation

② $R_2 = \{(1,1), (3,3)\}$ Only diagonal order pairs in R_2 , so No problem
Hence anti-symmetric

③ $R_3 = \{(1,1), (1,2), (3,2)\}$ It is an anti-symmetric Relⁿ

$(2,1) \notin R_3$
 $\therefore$ OK

$(2,3) \notin R_3$
 $\therefore$ OK

④ $R_4 = \{(1,2), (2,3), (3,1), (3,2)\}$

for "2 & 3"
both $(2,3) \in R_4$ & $(3,2) \in R_4$
but $(2 \neq 3)$ Hence R_4 is not an anti-symmetric relⁿ
then is false

if is true

Note:

If a relation does not contain any non-diagonal order pair, then every such relation is symmetric as well as anti-symmetric

→ Largest symmetric as well as anti-symmetric relation on set A is Δ_A
↑
(diagonal relation)

Q: Let $A = \{1, 2, 3, \dots, n\}$
How many "anti-symmetric relation" are possible on set A
where $|A| = n$.

Number of
Anti-symmetric
relation

(Any number of
diagonal order
Pairs may be
Present)

And

(Possible Combinations
w.r.t. non-diagonal
order Pairs)

(2^n)

*

}

?

}

Approach w.r.t. non-diagonal order pairs

→ In a non-diagonal order pair (a, b) , we know that $a \neq b$

if $a = b$, then (a, b) is a diagonal order pair.

• For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

→ W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation.

Case (i)

both $(a, b), (b, a)$
are present

Not Allowed in
Anti-Symmetric reln

Case (ii)

(a, b) is present
but (b, a) is not present

Allowed in
Anti-Symmetric reln

Case (iii)

(a, b) is not present
but (b, a) is present

Allowed in
Anti-Symmetric reln

Case (iv)

Both (a, b) and (b, a)
are not present

Allowed in
Anti-Symmetric reln

Let $A = \{a, b, c, d\}$

Out of '4' we can choose two distinct elements in ${}^4C_2 = 6$ ways

Total possible combinations w.r.t. non-diagonal order pairs in a symmetric Relⁿ

$a \& b$



'3'

Choices w.r.t. $a \& b$

And ~~*~~

$a \& c$



3

Order ~~*~~

$a \& d$



3

Order ~~*~~

$b \& c$



3

Order ~~*~~

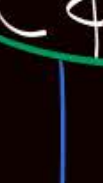
$b \& d$



3

Order ~~*~~

$c \& d$



3

$$= 3^6 = 3^{({}^4C_2)}$$

ie. $((a,b), (b,a))$ both absent
or
 $((a,b)$ present & (b,a) absent)
or
 $((a,b)$ absent & (b,a) present)

Q: Let $A = \{1, 2, 3, \dots, n\}$
 How many "anti-symmetric relation" are possible on set A
 where $|A| = n$.

Number of Anti-symmetric relation = (Any number of diagonal order Pairs may be Present) and (Possible combinations w.r.t. non-diagonal order Pairs)

$$(2^n)$$

$$3^{(nC_2)}$$

$$= 2^n \cdot 3^{\left(\frac{n^2-n}{2}\right)}$$



Topic : Asymmetric Relation

A relation R on set A is called asymmetric relation if and only if

if $a^R b$, then $b \not^R a$, $\forall a, b \in A$

i.e., if $(a, b) \in R$ then $(b, a) \notin R$, $\forall a, b \in A$

Does not matter whether $a = b$ or not

i.e. Diagonal order pairs are not allowed in asymmetric relⁿ

eg. let $A = \{1, 2, 3\}$

① $R_1 = \{ \}$ Smallest asymmetric relation on set A .

② $R_2 = \{(1, 2), (3, 3)\}$ diagonal not allowed $\therefore R_2$ is not an asymmetric Relⁿ

③ $R_3 = \{(1, 2), (2, 3), (3, 1), (3, 2)\}$ $\therefore$ Not an asymmetric Relⁿ

④ $R_4 = \{(\underline{1}, 2), (\underline{3}, 1), (\underline{3}, 2)\}$ R_4 is an asymmetric relⁿ

Q: Let $A = \{1, 2, 3, \dots, n\}$
How many "asymmetric relation" are possible on set A
where $|A| = n$.

Number of asymmetric relation = $\left(\begin{array}{l} \text{None of the} \\ \text{diagonal order} \\ \text{Pair is allowed} \end{array} \right)$ And $\left(\begin{array}{l} \text{Possible Combinations} \\ \text{w.r.t. Non-diagonal} \\ \text{Order Pairs} \end{array} \right)$

$\downarrow$ $\downarrow$ $\downarrow$

$\left({}^nC_0 \right)$ $\{ * \quad ? \}$

Approach w.r.t. non-diagonal order pairs

→ In a non-diagonal order pair (a, b) , we know that $a \neq b$

if $a = b$, then (a, b) is a diagonal order pair.

• For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

→ W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation.

Case (i)

both $(a, b), (b, a)$
are present

Not Allowed in
Asymmetric reln

Case (ii)

(a, b) is present
but (b, a) is not present

Allowed in
Asymmetric reln

Case (iii)

(a, b) is not present
but (b, a) is present

Allowed in
Asymmetric reln

Case (iv)

Both (a, b) and (b, a)
are not present

Allowed in
Asymmetric reln

Q: Let $A = \{1, 2, 3, \dots, n\}$
How many "asymmetric relation" are possible on set A
where $|A| = n$.

Number of
asymmetric
relation

(None of the
diagonal order
Pair is allowed)

And

(Possible Combinations
w.r.t. non-diagonal
order Pairs)

$\downarrow$
 $\binom{n}{0}$

*

$\downarrow$
 $3^{\binom{n}{2}}$

=

1

*

$3^{\frac{n^2-n}{2}}$

=

$3^{\frac{n(n-1)}{2}}$

Notes:

- ① On a set A , Empty relation is the only relation which is both symmetric as well as asymmetric
- ② Any subset of an anti-symmetric relation is also anti-symmetric
- ③ Every asymmetric relation is anti-symmetric as well, but Every anti-symmetric relation need not be an asymmetric relation

Note:

④

Any subset of asymmetric relation is also an asymmetric relation



Topic : Transitive Relation

A relation R on set A is called transitive relation if and only if

if $(a^R b \text{ and } b^R c)$ then $a^R c, \forall a, b, c \in A$

i.e. if $((a, b) \in R \text{ and } (b, c) \in R)$ then $(a, c) \in R, \forall a, b, c \in A$

H.W

eg:- Let $A = \{1, 2, 3\}$

① $R_1 = \{ \}$

② $R_2 = A \times A$

③ $R_3 = \{(1, 1), (2, 2)\}$

} check for
transitive relation

H.w

let $A = \{1, 2, 3\}$

④ $R_4 = \{ (1, 2), (2, 1), (1, 1) \}$

Check whether R_4 is
transitive or not

H.W.
eg.

$$A = \{1, 2, 3\}$$

$$R_5 = \{(1, 2), (1, 3)\}$$

transitive or not?

H.W.

eg:

$$A = \{1, 2, 3\}$$

$$R_6 = \{(2, 3), (1, 2)\}$$

transitive or not?

H.W

eg 1: Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$$

transitive or not?

H.W.

eg 2: Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$$

transitive or not?

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$

transitive or not?

eg. Let $A = \{a, b, c, d\}$

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

transitive or not?



2 mins Summary



✓ **Topic**

Cartesian product

✓ **Topic**

Relation

✓ **Topic**

Diagonal relation, Reflexive and irreflexive relation

✓ **Topic**

Symmetric, anti-symmetric and asymmetric relation

✓ **Topic**

Transitive relation

THANK - YOU